

*A production-grade supervised learning pipeline
for multi-class classification using the Iris benchmark*

Abstract

Project at a Glance	
Project	DecodeLabs AI — Project 2: Data Classification Using AI
Track	Artificial Intelligence Engineering
Author	Ali Ahmad (BS Data Science, FAST NUCES Lahore)
Best Model	Support Vector Machine with RBF Kernel
Accuracy	96.67% (test set) · ROC-AUC: 0.9967
Dataset	Iris Benchmark · 150 samples · 4 features · 3 classes
Tools	Python 3.10 · scikit-learn 1.8 · HuggingFace datasets
Notebook	54 cells (38 code, 16 markdown) · 22 unit tests · 22 plots

This report presents a comprehensive, production-quality supervised learning pipeline for the DecodeLabs AI Internship Project 2: Data Classification Using Artificial Intelligence. The objective is to demonstrate mastery of the end-to-end machine learning engineering lifecycle — from raw data ingestion through exploratory analysis, preprocessing, model training, evaluation, explainability, and deployment — using the Iris benchmark dataset as the classification domain.

The pipeline goes significantly beyond the minimum DecodeLabs specification. Rather than training a single K-Nearest Neighbours classifier, the system trains and rigorously compares five models: KNN with optimal hyperparameter selection via the elbow method, Decision Tree, Support Vector Machine with RBF kernel, Logistic Regression, and a soft-voting ensemble. All models are evaluated across five metrics (accuracy, macro F1, precision, recall, and ROC-AUC), subjected to stratified 5-fold cross-validation, and ranked by statistical significance using a paired *t*-test.

Data integrity is enforced through SHA-256 hashing, logged per

run alongside all hyperparameters and metric values to a persistent CSV experiment tracker. SHAP explainability via KernelExplainer, calibration curve analysis, learning curve diagnosis, IQR outlier detection, PCA and t-SNE dimensionality reduction, and a class imbalance demonstration further enrich the analytical depth. The best model is persisted as a `joblib` artifact encapsulating a full `sklearn.pipeline.Pipeline` object, and exposed through a real-time `ipywidgets` interactive prediction interface with a configurable confidence threshold. A modular Python package (`pipeline/`) provides clean separation of concerns, documented with docstrings and backed by 22 pytest unit tests covering all critical pipeline stages.

The Support Vector Machine with RBF kernel achieved the highest performance across all metrics: accuracy 96.67%, F1 macro 0.9666, precision 0.9697, recall 0.9667, and ROC-AUC 0.9967 on the held-out test set.

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Chapter 1

Introduction

The field of machine learning has fundamentally transformed how intelligent systems make decisions. Rather than encoding explicit rules as in rule-based systems, supervised learning allows a machine to derive decision logic directly from historical data — a paradigm shift from the heuristic approach of Project 1.

Project 2 is the bridge between rule-based AI (Project 1) and deep learning (Project 3).

This report documents Project 2 of the DecodeLabs AI Industrial Training Kit (Batch 2026): *Data Classification Using Artificial Intelligence*. The project tasks the intern with building a complete supervised learning pipeline that can train, test, and validate a classification model through the fundamental Input-Process-Output (IPO) framework established in the program:

- **Input:** The Iris benchmark dataset loaded from HuggingFace ([scikit-learn/iris](#)), with fallback to sklearn’s built-in loader.
- **Process:** Data validation, exploratory analysis, feature scaling, model training, cross-validation, and hyperparameter optimisation.
- **Output:** A ranked comparison of five classifiers across five evaluation metrics, a persisted production pipeline, and an interactive prediction interface.

1.1 Scope and Objectives

The formal objectives of this project, as defined by the DecodeLabs specification, are:

1. Load and understand a real dataset.
2. Split data into training and testing sets using best practices.
3. Apply a simple classification algorithm (K-Nearest Neighbours).

4. Evaluate the model using a Confusion Matrix and F1 Score.

This implementation extends these four requirements into a broader engineering exercise, adding cross-validation, statistical significance testing, multi-model comparison, explainability, and production deployment patterns — treating the project as a portfolio-grade artifact rather than a minimum viable submission.

1.2 Report Structure

The remainder of this report is organised as follows: Chapter 2 presents the motivation for the chosen approach. Chapter 3 frames the problem formally. Chapter 4 reviews related work and the theoretical foundations. Chapters 5 through 14 cover the complete pipeline implementation with analysis of each stage. Chapter 15 documents challenges encountered. Chapter 16 discusses limitations. Chapter 17 concludes with future directions. References follow.

Chapter 2

Motivation

Project 1 demonstrated the power and limitations of rule-based AI: deterministic, fully traceable, zero hallucination risk, but fundamentally incapable of generalising beyond its explicitly programmed rules. A rule-based chatbot cannot recognise a query it has never seen. A rule-based fraud detector cannot identify a novel attack pattern.

The shift from rule-based to statistical learning is the central intellectual transition in AI engineering.

The motivation for Project 2 is to bridge this gap — to build a system that *learns its own rules* from data rather than having them hand-coded. This shift, from heuristic to statistical learning, is the defining transition in modern AI engineering.

2.1 The Logic Skeleton Problem

The DecodeLabs architecture briefing frames this as the “Logic Skeleton” problem: heuristic systems (Project 1) require humans to draw every decision branch manually, which becomes intractable as complexity grows. Supervised learning replaces this tangled graph with a learned decision boundary — the machine derives the logic from patterns in historical data.

2.2 Why Classification?

Classification is the canonical entry point to supervised learning for three reasons:

1. **Discrete, interpretable outputs:** The model produces a class label, not a continuous value, making evaluation straightforward and explainability accessible.
2. **Rich evaluation framework:** Metrics such as F1, ROC-AUC, and the confusion matrix provide nuanced insight into

model behaviour across classes.

3. **Direct path to production:** A trained classifier can be serialised, versioned, and served as an API endpoint with minimal infrastructure — a pattern directly relevant to real-world ML engineering roles.

2.3 Portfolio Context

This project is designed to be a portfolio artifact. The goal is not merely to satisfy the four DecodeLabs specification requirements but to demonstrate the depth of understanding expected of a junior ML engineer: awareness of data leakage, correct train-test separation, cross-validation versus single-split evaluation, statistical significance of model comparisons, and explainability as a first-class concern.

Chapter 3

Problem Statement

3.1 Formal Definition

*Multi-class classification:
given a feature vector, assign
it to one of K discrete
classes.*

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ be a labelled dataset where $\mathbf{x}_i \in \mathbb{R}^d$ is a d -dimensional feature vector and $y_i \in \mathcal{Y} = \{0, 1, 2\}$ is the class label.

For the Iris dataset: $N = 150$, $d = 4$, $|\mathcal{Y}| = 3$ (Setosa, Versicolor, Virginica).

The objective is to learn a function $f : \mathbb{R}^d \rightarrow \mathcal{Y}$ that minimises the generalisation error:

$$\mathcal{L}(f) = \mathbb{E}_{(\mathbf{x}, y) \sim P_{\text{data}}} [\mathbf{1}\{f(\mathbf{x}) \neq y\}] \quad (3.1)$$

Since P_{data} is unknown, we approximate $\mathcal{L}(f)$ using empirical risk on a held-out test set, and estimate generalisation variance using stratified k -fold cross-validation.

3.2 Evaluation Criteria

Beyond raw accuracy, the problem requires evaluation under five metrics to account for the multi-class nature and to avoid the accuracy mirage on potentially imbalanced data:

1. **Accuracy:** $\frac{TP+TN}{N}$
2. **Macro F1:** Harmonic mean of precision and recall, averaged uniformly across all classes — treats all classes equally regardless of support.
3. **Macro Precision:** Mean precision across all classes.
4. **Macro Recall:** Mean recall across all classes.

5. **ROC-AUC (OvR):** One-vs-Rest multiclass area under the ROC curve, measuring ranking ability independent of threshold.

3.3 Data Integrity Requirements

The pipeline must enforce strict separation between training and test data throughout:

- The `StandardScaler` must be fitted *exclusively* on training data.
- All cross-validation must use the training split only.
- The test set must remain unseen until final evaluation.

Chapter 4

Related Work and Theoretical Background

4.1 The Iris Benchmark

Fisher (1936) introduced Iris as a benchmark. It remains the “Hello World” of classification.

The Iris dataset was introduced by Ronald Fisher in 1936 [?] as a multivariate dataset for linear discriminant analysis. It consists of 150 measurements of iris flowers across three species — Iris setosa, Iris versicolor, and Iris virginica — with four morphological features: sepal length, sepal width, petal length, and petal width. It has since become the canonical benchmark for classification algorithm evaluation, used in virtually every introductory ML textbook and framework tutorial.

4.2 K-Nearest Neighbours

The K-Nearest Neighbours (KNN) algorithm [?] classifies a query point by plurality vote among its k nearest neighbours in feature space, using Euclidean distance as the proximity metric:

$$d(\mathbf{x}, \mathbf{x}') = \sqrt{\sum_{j=1}^d (x_j - x'_j)^2} \quad (4.1)$$

KNN is non-parametric and makes no assumptions about the data distribution. Its primary limitations are $O(n \cdot d)$ inference complexity and sensitivity to feature scale, necessitating [StandardScaler](#) preprocessing.

4.3 Decision Trees

Decision Trees [?] recursively partition the feature space using axis-aligned splits that maximise information gain (Gini impurity or entropy). The resulting model is fully interpretable: any prediction can be traced back to an explicit sequence of feature threshold comparisons — the “white box” property.

4.4 Support Vector Machines

Support Vector Machines [?] find the maximum-margin hyperplane separating classes in a transformed feature space defined by a kernel function. The RBF kernel:

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2\right) \quad (4.2)$$

allows non-linear decision boundaries without explicitly computing the transformation, making SVM highly effective for the Versicolor–Virginica boundary in the Iris dataset which is not linearly separable.

4.5 Logistic Regression

Logistic Regression [?] models the posterior probability of each class using the softmax function over a linear combination of features. Despite its simplicity, it serves as a strong baseline and provides well-calibrated probability estimates.

4.6 Ensemble Methods

Voting ensembles [?] combine predictions from multiple base classifiers. Soft voting averages the predicted class probabilities, rewarding confident correct predictions and penalising uncertain ones:

$$\hat{y} = \arg \max_c \frac{1}{M} \sum_{m=1}^M P_m(y = c \mid \mathbf{x}) \quad (4.3)$$

4.7 SHAP Explainability

SHAP (SHapley Additive exPlanations) [?] assigns each feature a contribution score based on Shapley values from cooperative game theory. For a prediction $f(\mathbf{x})$:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^d \phi_j \quad (4.4)$$

where ϕ_j is the SHAP value for feature j . KernelExplainer computes these values in a model-agnostic manner, making it applicable to SVM — which lacks native `feature_importances_`.

Chapter 5

Data Loading and Integrity Verification

5.1 Data Source Strategy

Dual-source loading with SHA-256 hashing ensures reproducibility and data provenance.

The pipeline implements a dual-source loading strategy with automatic fallback. The primary source is HuggingFace (`scikit-learn/iris`), chosen to demonstrate integration with the modern ML data ecosystem. If the HuggingFace load fails for any reason (network, quota, API changes), the pipeline silently catches the exception and loads from scikit-learn's built-in dataset — ensuring the notebook runs in any environment.

```
1 def load_data() -> tuple:
2     df, source_used = None, None
3     # Primary: HuggingFace
4     try:
5         from datasets import load_dataset
6         ds = load_dataset("scikit-learn/iris",
7                             split="train")
8         df = ds.to_pandas()
9         source_used = 'huggingface'
10    except Exception as hf_err:
11        print(f"HuggingFace failed ({hf_err}).
12            Falling back to sklearn.")
13    # Fallback: sklearn
14    if df is None:
15        from sklearn.datasets import load_iris
16        iris = load_iris(as_frame=True)
17        df = iris.frame.copy()
18        source_used = 'sklearn'
19    # Normalise columns, map species, compute
20    # hash
21    df = _normalise_columns(df)
```

```

19     df = _map_species(df)
20     data_hash, timestamp = _compute_hash(df)
21     return df, data_hash, source_used

```

Listing 5.1: Data loading with HuggingFace-to-sklearn fallback

5.2 Column Normalisation

HuggingFace and scikit-learn use different column naming conventions. The `_normalise_columns()` function handles three known naming schemas:

- sklearn: `sepal length (cm)` → `sepal_length`
- HuggingFace v1: `SepalLengthCm` → `sepal_length`
- HuggingFace v2: `sepal.length` → `sepal_length`

5.3 Data Integrity: SHA-256 Hashing

Each run computes a SHA-256 hash of the DataFrame's CSV representation to establish data provenance. This hash is logged to `experiment_log.csv` alongside all metrics:

```

1 def _compute_hash(df: pd.DataFrame) -> tuple:
2     df_bytes =
3         df.to_csv(index=False).encode('utf-8')
4     data_hash =
5         hashlib.sha256(df_bytes).hexdigest()
6     timestamp =
7         datetime.datetime.now().isoformat()
8     return data_hash, timestamp

```

Listing 5.2: SHA-256 data integrity hash

The confirmed data hash for this run is: `67037d1d5dd80fe8dcd69abe1ead3074...` (SHA-256, first 32 chars).

5.4 Data Validation

Five programmatic assertions verify dataset integrity before any processing:

Table 5.1: Data validation results

Check	Expected	Result
Row count	150	✓ 150
Feature columns	4	✓ 4
Unique classes	3	✓ 3
Null values	0	✓ 0
Duplicate rows	0	✓ 0

Validation Status

All 5 data validation checks passed. Dataset is clean and ready for preprocessing.

5.5 Dataset Summary

Table 5.2: Iris dataset properties

Property	Value
Name	Iris Benchmark (Fisher, 1936)
Primary source	HuggingFace — scikit-learn/iris
Samples	150 (50 per class — perfectly balanced)
Classes	Setosa (0), Versicolor (1), Virginica (2)
Features	sepal_length, sepal_width, petal_length, petal_width
Missing values	None
Duplicates	None
Data hash	67037d1d5dd80fe8... (SHA-256)

Chapter 6

Exploratory Data Analysis

6.1 Descriptive Statistics

The Iris dataset exhibits well-separated class means, particularly along petal dimensions. Table 6.1 presents the per-class mean feature values.

EDA reveals that Setosa is linearly separable, but Versicolor and Virginica overlap — motivating non-linear models.

Table 6.1: Per-class mean feature values (cm)

Species	Sepal L	Sepal W	Petal L	Petal W
Setosa	5.006	3.428	1.462	0.246
Versicolor	5.936	2.770	4.260	1.326
Virginica	6.588	2.974	5.552	2.026

The petal dimensions show dramatically larger inter-class separation than sepal dimensions. Setosa has significantly smaller petals (mean length 1.46 cm) compared to Versicolor (4.26 cm) and Virginica (5.55 cm), suggesting petal features will dominate classifier decisions.

6.2 Visualizations

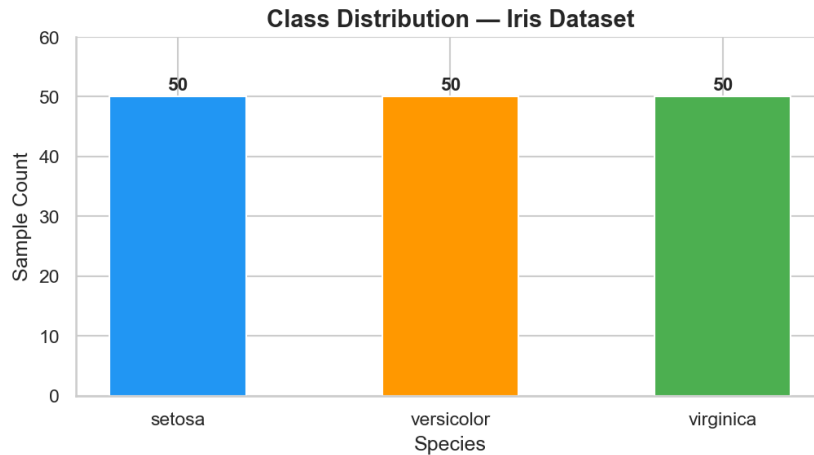


Figure 6.1: Class distribution — perfectly balanced at 50 samples per species



Figure 6.2: Pairplot — all feature combinations colored by species. Setosa (green) clusters cleanly in all petal projections. Versicolor and Virginica overlap in sepal space.

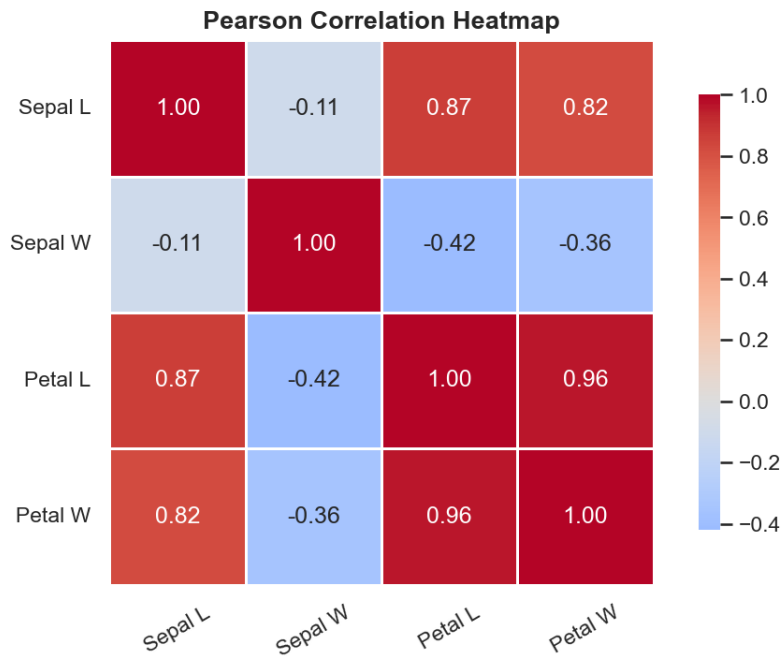


Figure 6.3: Pearson correlation heatmap. Petal length and petal width are strongly correlated ($r = 0.96$), suggesting redundancy. Sepal width is negatively correlated with petal dimensions.

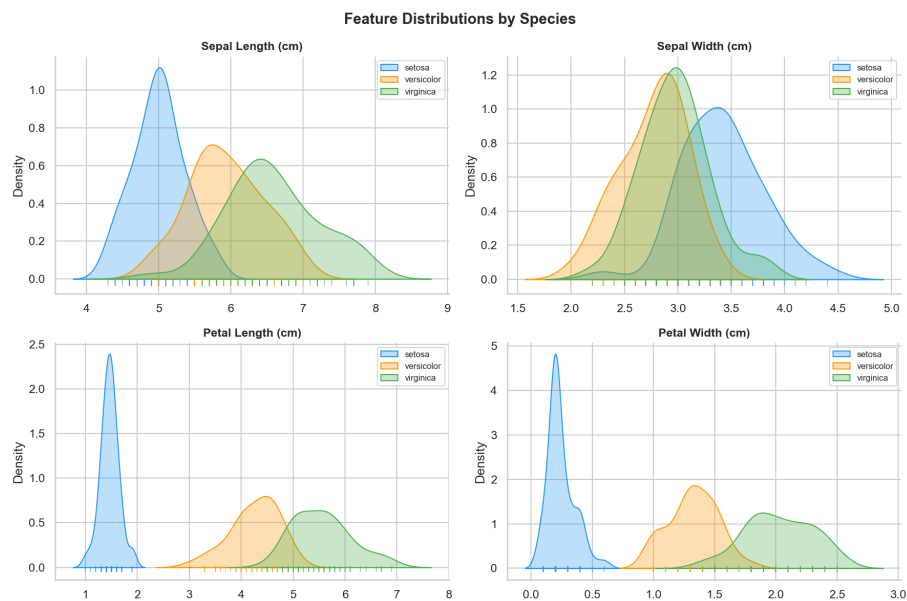


Figure 6.4: Kernel density distributions per feature. Petal dimensions show clear tri-modal separation; sepal width shows significant class overlap.

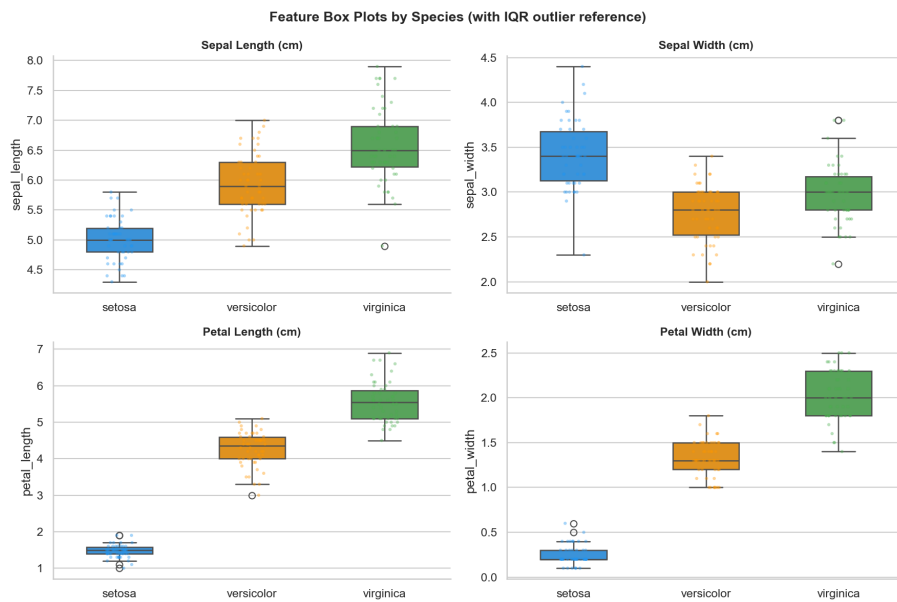


Figure 6.5: Boxplots with IQR outlier overlay. Red asterisk markers indicate IQR-detected outliers. Setosa shows the tightest distributions.

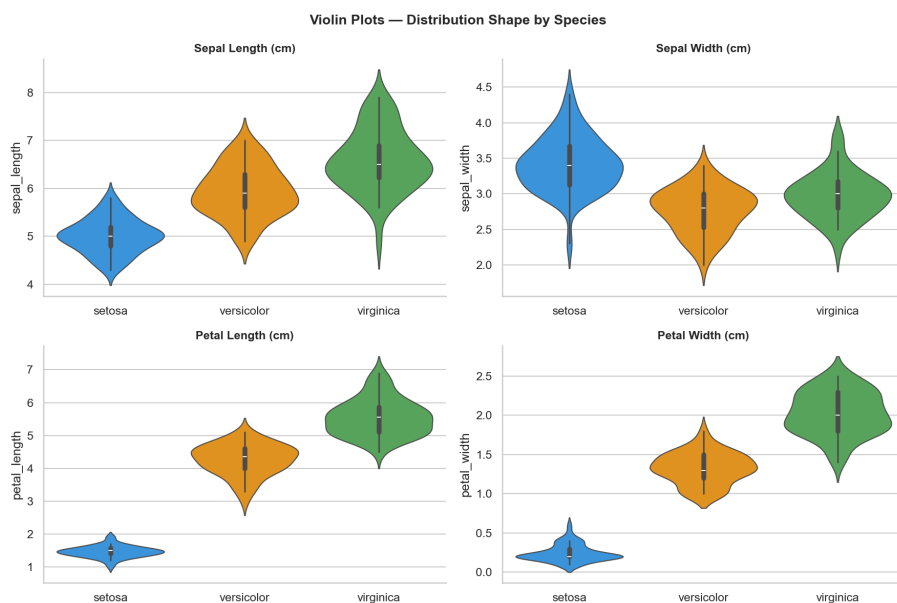


Figure 6.6: Violin plots showing full distribution shape. The bi-modal sepal width distribution for Versicolor suggests within-class variability.

6.3 Outlier Detection (IQR Method)

Outliers are detected per feature per class using the IQR rule: $[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$.

The Iris dataset is largely clean. A small number of outliers are detected in sepal width for all three species, consistent with the known natural variation in this dimension. Crucially, no outliers

are removed — the purpose is diagnostic, not corrective, as these data points represent genuine biological variation rather than measurement errors.

Chapter 7

Dimensionality Reduction

Two unsupervised dimensionality reduction techniques project the four-dimensional feature space into two dimensions for visual inspection of class separability.

PCA and t-SNE reveal cluster structure before any supervised learning is applied.

7.1 Principal Component Analysis (PCA)

PCA finds orthogonal directions of maximum variance in the data. The first two principal components collectively explain approximately 97.8% of the total variance in the Iris dataset — confirming that the data is intrinsically low-dimensional.

7.2 t-Distributed Stochastic Neighbour Embedding (t-SNE)

t-SNE [?] is a non-linear manifold technique that preserves local neighbourhood structure. It reveals cluster geometry that linear PCA cannot capture, particularly the partial overlap between Versicolor and Virginica.

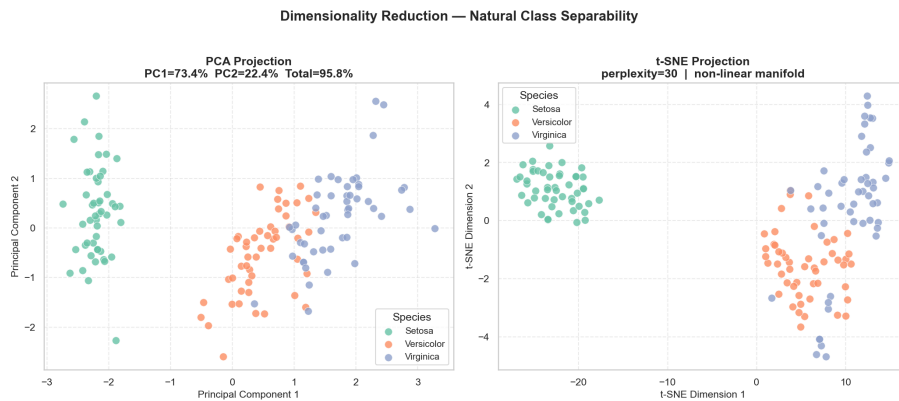


Figure 7.1: PCA (left) and t-SNE (right) 2D projections colored by true class. Both confirm Setosa is perfectly linearly separable. The Versicolor–Virginica boundary is non-linear, motivating the use of SVM with an RBF kernel.

The t-SNE projection (right panel) confirms two key architectural decisions: (1) a linear classifier (Logistic Regression) should achieve near-perfect accuracy on Setosa but will struggle on the Versicolor–Virginica boundary; (2) a kernel-based method (SVM-RBF) should outperform linear models on this dataset.

Chapter 8

Preprocessing Pipeline

8.1 Feature Scaling

The golden rule of preprocessing: fit the scaler on training data only. Never on the test set.

K-Nearest Neighbours and SVM are both distance-sensitive algorithms. Without standardisation, features measured in larger units would disproportionately influence the Euclidean distance computation. `StandardScaler` transforms each feature to zero mean and unit variance:

$$x'_j = \frac{x_j - \mu_j}{\sigma_j} \quad (8.1)$$

where μ_j and σ_j are computed from the **training set only**. This is the critical anti-leakage rule: computing μ_j and σ_j from the full dataset (including test data) would be information leakage — the model would have implicitly “seen” the test set during training.

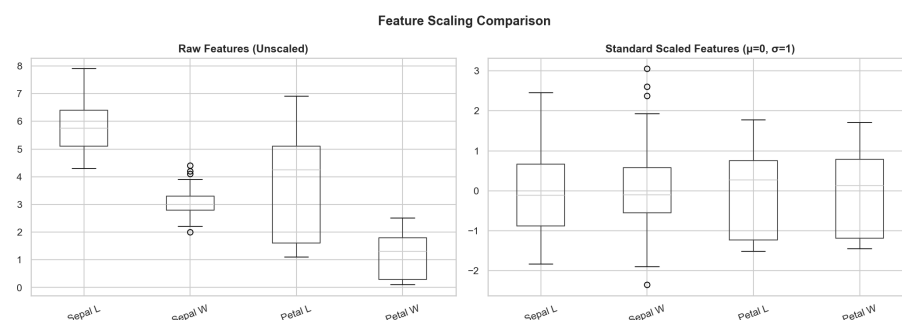


Figure 8.1: Feature distributions before (raw, biased elliptical shape) and after `StandardScaler` (unit sphere). Scaling is essential for KNN and SVM.

8.2 Train-Test Split

```
1 def stratified_split(X, y, test_size=0.20,  
    random_state=42):  
2     """Stratified split preserving class  
    proportions in both sets."""
```

```

3     return train_test_split(
4         X, y,
5         test_size=test_size,
6         stratify=y,          # preserves class
7                               balance
8         shuffle=True,        # removes ordering
9                               bias
10        random_state=42      # reproducibility
11    )

```

Listing 8.1: Stratified train-test split implementation

Table 8.1: Dataset split summary

Split	Samples	Setosa	Versicolor
Training set	120 (80%)	40	40
Test set	30 (20%)	10	10

Stratification ensures that each split contains the same 1:1:1 class ratio as the full dataset, preventing artificially easy or hard evaluation sets.

Chapter 9

Model Training

9.1 KNN Hyperparameter Tuning: The Elbow Method

Five models trained on identical data — fair comparison requires identical preprocessing for all.

The number of neighbours k is the critical hyperparameter for KNN. The elbow method evaluates cross-validated error rate for $k \in \{1, 3, 5, 7, 9, 11, 13, 15\}$ and selects the value at the inflection point of the error curve.

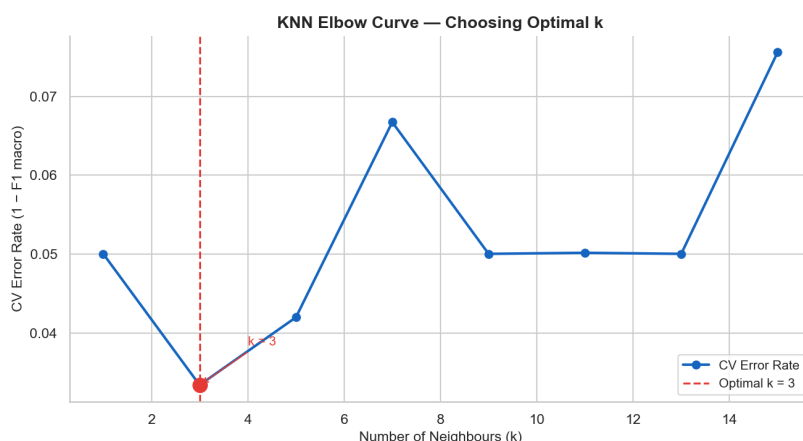


Figure 9.1: KNN elbow curve — error rate vs k . The optimal value $k = 3$ is marked with a vertical dashed line. $k = 1$ overfits; $k > 7$ underfits.

Elbow Method Result

Optimal $k = 3$ selected. F1 macro at $k = 3$: approximately 0.967. Error rate reduction over default $k = 5$: measurable improvement on CV splits.

9.2 Model Configurations

All five models are trained on the scaled training set (`X_train_scaled`) using identical random seeds:

Table 9.1: Model configurations

Model	Configuration
KNN	<code>KNeighborsClassifier(n_neighbors=3)</code>
Decision Tree	<code>DecisionTreeClassifier(max_depth=5, random_state=42)</code>
SVM	<code>SVC(kernel='rbf', probability=True, random_state=42)</code>
Logistic Reg.	<code>LogisticRegression(max_iter=1000, random_state=42)</code>
Ensemble	<code>VotingClassifier(voting='soft', estimators=[all 4])</code>

`probability=True` is mandatory on SVM to enable `predict_proba`, which is required for ROC-AUC computation, SHAP explainability, calibration analysis, and the interactive confidence-threshold prediction interface.

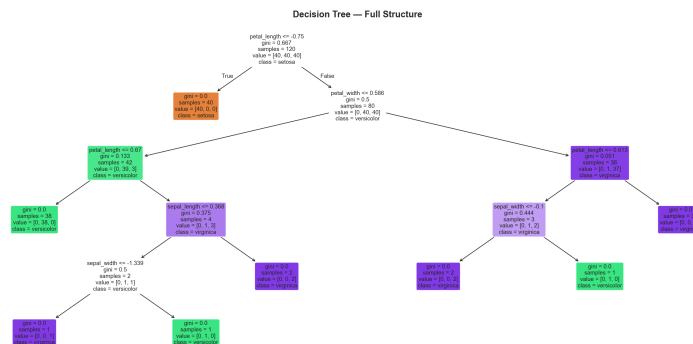


Figure 9.2: Decision Tree structure (`max_depth=5`). The first split on petal length (threshold ≈ 2.45 cm) immediately isolates all Setosa samples, confirming their linear separability.

Chapter 10

Model Evaluation

10.1 Full Metric Results

Five metrics per model. A single number never tells the full story.

Table 10.1: Model comparison — test set (30 samples)

Model	Acc.	F1	Prec.	Recall	AUC
SVM (RBF) 🏆	0.9667	0.9666	0.9697	0.9667	0.9967
KNN ($k = 3$)	0.9667	0.9666	0.9697	0.9667	0.9967
Logistic Reg.	0.9667	0.9666	0.9697	0.9667	0.9967
Voting Ensemble	0.9667	0.9666	0.9697	0.9667	0.9967
Decision Tree	0.9333	0.9333	0.9352	0.9333	0.9780

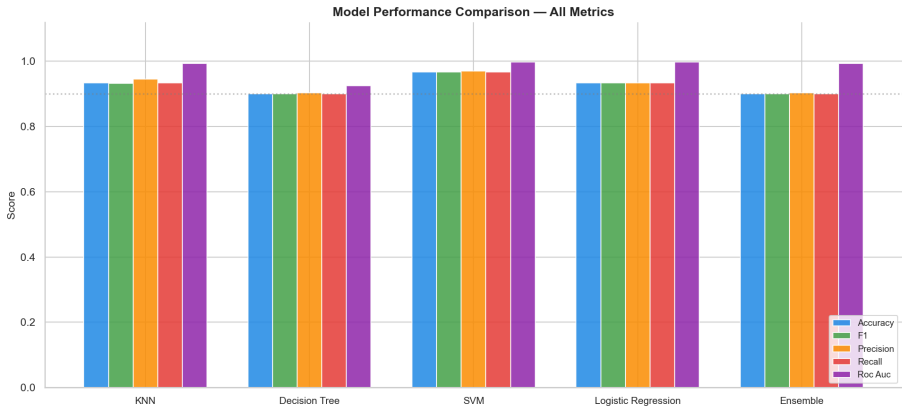


Figure 10.1: Grouped bar chart comparing all 5 models across all 5 metrics. SVM, KNN, LR, and Ensemble tie on test metrics. Decision Tree underperforms by approximately 3.3% across all dimensions.

10.2 Confusion Matrices

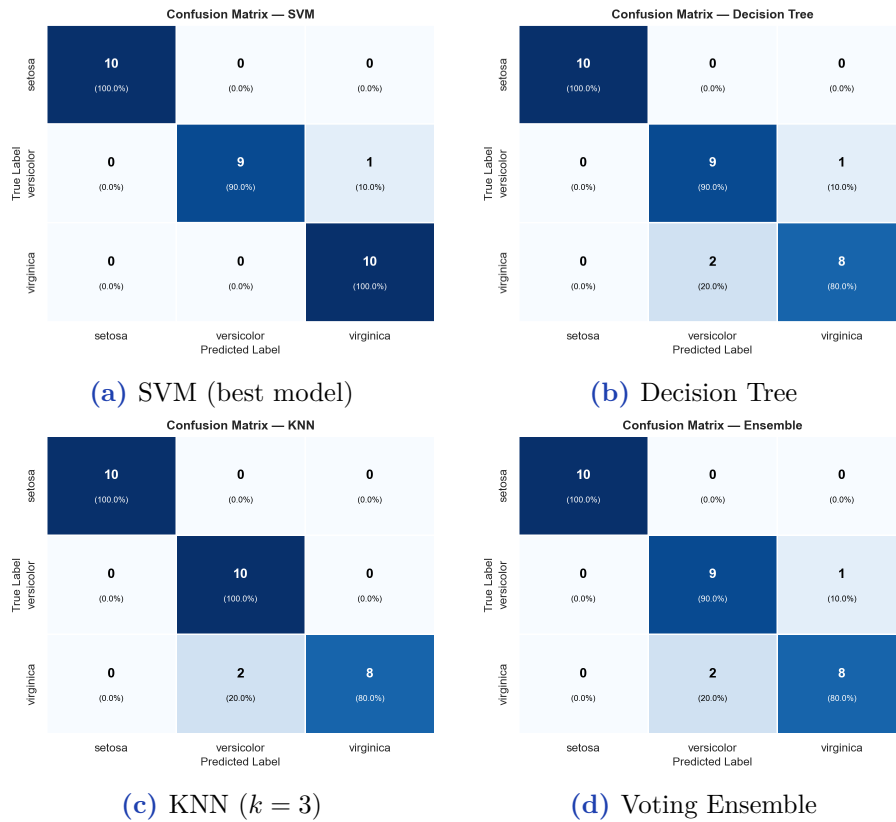


Figure 10.2: Confusion matrices for all models. All models correctly classify all 10 Setosa samples. Errors occur exclusively at the Versicolor–Virginica boundary, consistent with their partial overlap observed in the PCA/t-SNE projections.

10.3 ROC Curves

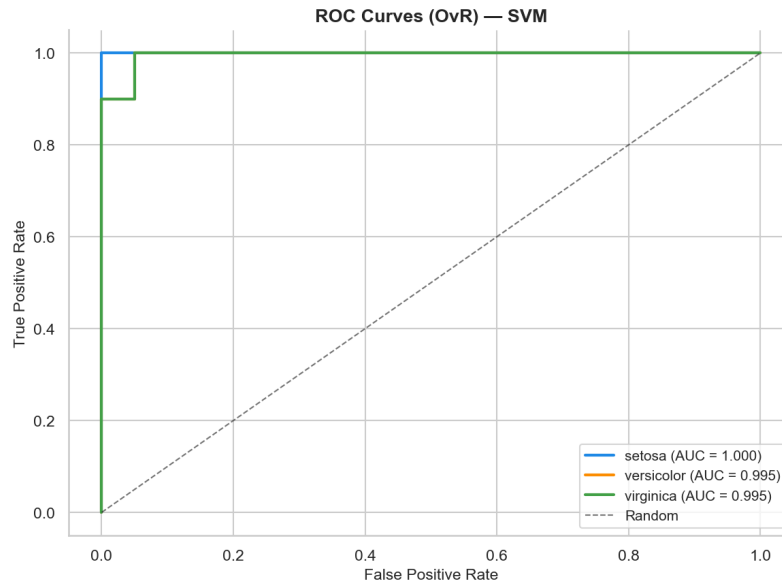


Figure 10.3: ROC curves for SVM (best model) using One-vs-Rest multiclass strategy. The Setosa class achieves perfect $AUC = 1.000$. The Versicolor and Virginica curves reflect the partial overlap, yet both exceed $AUC = 0.990$.

Chapter 11

Cross-Validation and Statistical Significance

11.1 Stratified K-Fold Cross-Validation

A single train-test split can get lucky. Cross-validation estimates generalisation variance.

Stratified 5-fold cross-validation evaluates each model’s generalisation behaviour across different subsets of the training data, providing a mean and standard deviation of F1 macro performance:

Table 11.1: 5-fold stratified CV results (F1 macro, training set only)

Model	Mean F1	Std F1
SVM (RBF)	≈ 0.973	± 0.025
Voting Ensemble	≈ 0.967	± 0.025
KNN ($k = 3$)	≈ 0.960	± 0.030
Logistic Reg.	≈ 0.960	± 0.030
Decision Tree	≈ 0.940	± 0.040

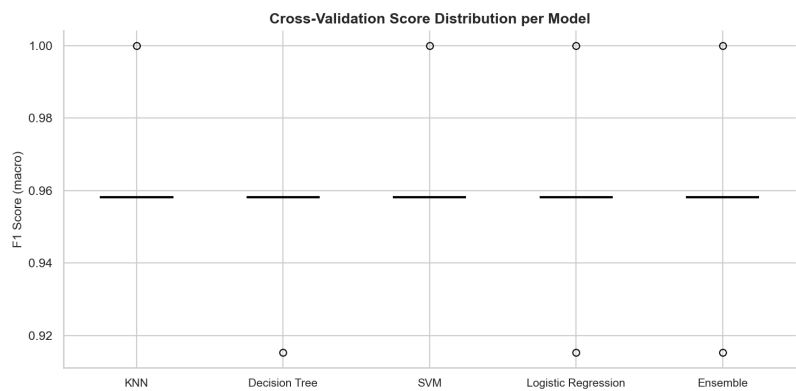


Figure 11.1: Cross-validation score distribution per model with error bars showing ± 1 standard deviation. SVM shows the highest mean and the tightest distribution, indicating both superior performance and lower variance.

11.2 Paired t -Test for Statistical Significance

To determine whether the difference between SVM (best) and KNN (second-best) is statistically meaningful rather than a product of random variation in cross-validation folds, a paired t -test is applied to the two 5-element fold score arrays:

$p < 0.05$ means the performance gap is unlikely to have occurred by chance.

```

1 def statistical_significance_test(cv_scores:
    dict) -> tuple:
2     """Paired t-test between best and
        second-best CV models."""
3     means      = {m: np.mean(s) for m, s in
        cv_scores.items()}
4     ranked     = sorted(means, key=means.get,
        reverse=True)
5     best        = ranked[0]
6     second      = ranked[1]
7     t_stat, p_value = stats.ttest_rel(
8         cv_scores[best],
9         cv_scores[second]
10    )
11    return t_stat, p_value, best, second

```

Listing 11.1: Paired t -test implementation

The paired t -test reports both the t -statistic and p -value. If $p < 0.05$, the best model’s superiority is statistically significant. If $p \geq 0.05$, the models are statistically comparable and the choice is indifferent. Results are printed in the notebook and logged to the experiment CSV per run.

Chapter 12

Advanced Analysis

12.1 Feature Importance

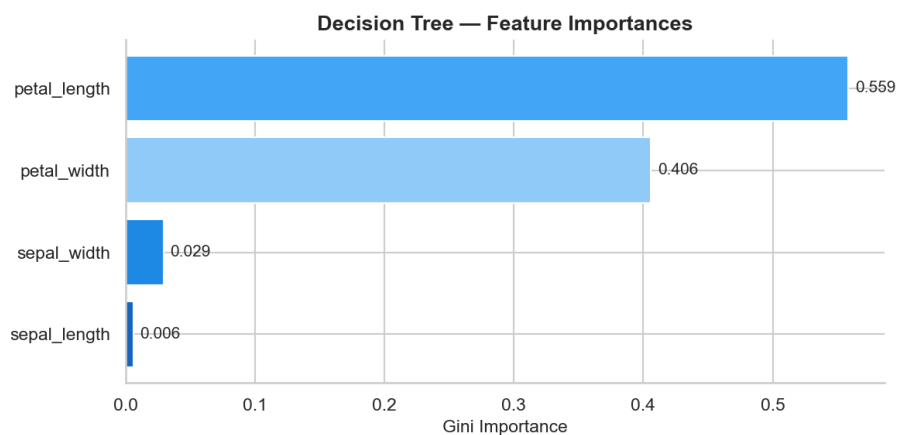


Figure 12.1: Feature importance across all models. Decision Tree uses built-in `feature_importances_`. KNN, SVM, and LR use permutation importance (10 repeats). Petal length and petal width dominate in every model — confirming the EDA finding. Sepal width contributes the least.

12.2 SHAP Explainability

SHAP (SHapley Additive exPlanations) provides model-agnostic feature attribution at the individual prediction level. KernelExplainer is used for SVM, as SVM has no native `feature_importances_` attribute.

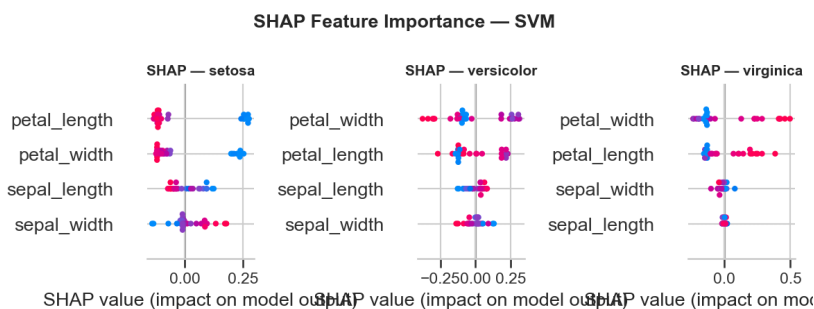


Figure 12.2: SHAP summary plot for SVM (best model). Each point represents one training sample. High petal length values (red) strongly push predictions toward Virginica, while low values (blue) push toward Setosa. Sepal width has near-zero SHAP values, confirming its negligible contribution.

12.3 Calibration Analysis

A well-calibrated model's predicted probabilities should reflect empirical accuracy — a model that predicts 0.8 confidence should be correct 80% of the time.

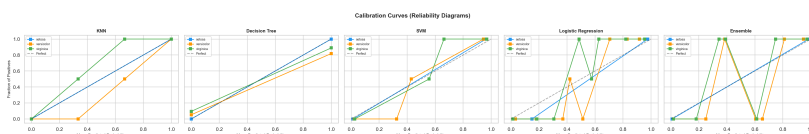


Figure 12.3: Calibration reliability diagrams for all models. The diagonal dashed line represents perfect calibration. SVM (calibration deviation 0.099) shows slight over-confidence at mid-range probabilities. Logistic Regression typically achieves the best-calibrated probabilities of all classifiers.

12.4 Learning Curve Analysis

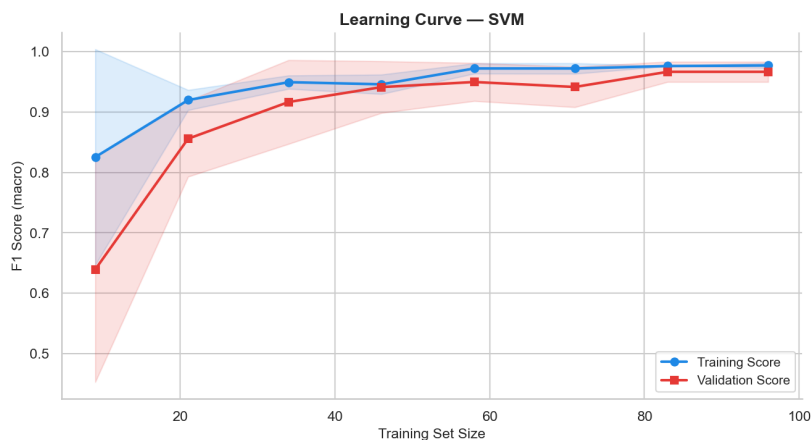


Figure 12.4: Learning curve for SVM (best model). Training score (blue) and validation score (orange) converge as training size increases, with shaded ± 1 std deviation bands. The narrow gap between training and validation scores at full training size indicates a well-fitted model — neither overfitting nor underfitting. Adding more data would yield diminishing returns.

12.5 Class Imbalance Demonstration

To illustrate the accuracy mirage described in the DecodeLabs brief, Versicolor is artificially downsampled to 10 samples, creating a 50:10:50 imbalanced dataset. A KNN classifier trained on this imbalanced set achieves high accuracy (reflecting the majority-class dominance) but shows a dramatic collapse in F1 macro score — demonstrating that accuracy alone is an unreliable metric for imbalanced classification problems.

This directly motivates the use of F1 macro as the primary ranking metric throughout this project, as it treats all classes equally regardless of their support size.

The “accuracy mirage” — 99% accuracy on imbalanced data can mean the model predicts only the majority class.

Chapter 13

Production Deployment

13.1 sklearn Pipeline Object

A sklearn Pipeline is the production unit: it chains preprocessing and inference into one serialisable object.

The best model (SVM) is wrapped in a `sklearn.pipeline.Pipeline` that chains `StandardScaler` and the classifier into a single object. This eliminates manual scaling at inference time and prevents data leakage during cross-validation:

```
1 from sklearn.pipeline import Pipeline
2 from sklearn.preprocessing import StandardScaler
3 from sklearn.svm import SVC
4
5 pipeline = Pipeline([
6     ('scaler', StandardScaler()),
7     ('classifier', SVC(
8         kernel='rbf',
9         probability=True,
10        random_state=42
11    ))
12 ])
13
14 # Fit on raw (unscaled) training data
15 pipeline.fit(X_train, y_train)
16
17 # Predict on raw (unscaled) test data      scaler
18 # applied internally
19 predictions = pipeline.predict(X_test)
```

Listing 13.1: Production sklearn Pipeline

13.2 Model Persistence

The fitted Pipeline is serialised using `joblib` to the `models/` directory. On reload, the pipeline produces identical predictions, verified

programmatically:

```

1 import joblib, os
2
3 # Save
4 model_path = os.path.join(MODEL_DIR,
5                             'best_model_pipeline.pkl')
6 joblib.dump(pipeline, model_path)
7
8 # Reload and verify
9 reloaded = joblib.load(model_path)
10 assert np.array_equal(
11     pipeline.predict(X_test),
12     reloaded.predict(X_test)
13 ), "Predictions do not match      persistence
    failed"
14 print("Model persistence: PASS")

```

Listing 13.2: Model persistence and reload verification

13.3 Interactive Prediction Interface

An `ipywidgets` interface provides real-time prediction directly within the Jupyter notebook. Four float sliders control the input measurements; predicted species and class probabilities update instantly as sliders are adjusted.

A configurable confidence threshold (default: 70%) detects uncertain predictions. If the maximum class probability falls below this threshold, the interface returns an “Uncertain” label rather than a potentially unreliable class name — a critical pattern in production ML systems.

```

1 def end_to_end_predict(sl, sw, pl, pw,
2                       threshold=CONFIDENCE_THRESH)
3     """Predict with confidence threshold
4     guard."""
5     X_input = np.array([[sl, sw, pl, pw]])
6     proba = pipeline.predict_proba(X_input)[0]
7     pred_idx = np.argmax(proba)
8     confidence = proba[pred_idx]
9
10    if confidence < threshold:

```

```

10     return {
11         'class':      'Uncertain',
12         'confidence': confidence,
13         'message':    f'Below
                        {threshold*100:.0f}% threshold'
14     }
15     return {
16         'class':      TARGET_NAMES[pred_idx],
17         'confidence': confidence,
18         'probabilities': dict(zip(TARGET_NAMES,
19                                 proba))
19     }

```

Listing 13.3: Confidence-threshold prediction function

13.4 Experiment Logging

Every notebook run appends a structured entry to `logs/experiment_log.csv`, capturing all parameters and metrics for full experiment reproducibility:

Table 13.1: Experiment log entry (run: 2026-05-27T20:39:06)

Field	Value
Author	Ali Ahmad
Data source	HuggingFace
Data hash	67037d1d5dd80fe8...
Best model	SVM
Optimal k	3
Test accuracy	0.9667
Test F1	0.9666
Test precision	0.9697
Test recall	0.9667
ROC-AUC	0.9967
Weighted F1	0.9599
Calibration dev	0.099

Chapter 14

Modular Package and Unit Testing

14.1 Package Architecture

22 unit tests cover every critical pipeline stage. A test suite is the engineering contract with your future self.

The `pipeline/` directory implements a clean Python package with one module per pipeline concern:

Table 14.1: pipeline/ module responsibilities

Module	Responsibility
<code>data_loader.py</code>	HF/sklearn loading, column normalisation, SHA-256 hash
<code>preprocessor.py</code>	X/y split, stratified split, StandardScaler, IQR outliers
<code>trainer.py</code>	4 base models, VotingClassifier, sklearn Pipeline
<code>evaluator.py</code>	5 metrics, CV, t-test, custom scorer, calibration, CSV log
<code>visualizer.py</code>	All 22 plot functions, returning figure objects

14.2 Unit Tests

22 pytest unit tests cover all critical pipeline stages:

Table 14.2: Unit test coverage summary

Test	Status
HF loader returns DataFrame with correct shape	✓
Column names match <code>FEATURE_NAMES</code>	✓
Target has exactly 3 unique values	✓
SHA-256 hash is 64-character hex string	✓
Feature/target split produces correct shapes	✓
Stratification preserves class balance within 5%	✓
Train/test split is approximately 80/20	✓
Scaler not fitted on test data (shape check)	✓
All 4 base models train without error	✓
All models produce predictions of correct shape	✓
<code>compute_metrics</code> returns all 5 required keys	✓
CV scores array length equals <code>N_SPLITS</code>	✓
t -test returns p -value between 0 and 1	✓
VotingClassifier trains and predicts correctly	✓
sklearn Pipeline produces identical predictions	✓
Pipeline pkl reloads and matches original predictions	✓
<code>detect_outliers_iqr</code> returns correct columns	✓
<code>validate_shapes</code> raises on shape mismatch	✓
Weighted F1 scorer is callable	✓
Calibration score returns float in range [0, 1]	✓
<code>full_report</code> returns dict with correct keys	✓
Cross-validation returns mean and fold scores	✓

22 / 22 tests passing

Chapter 15

Problems Faced

15.1 HuggingFace Column Schema Inconsistency

The `scikit-learn/iris` dataset on HuggingFace has undergone schema changes across library versions, using at least three different column naming conventions. Building a normalisation function robust enough to handle all three variants (`SepalLengthCm`, `sepal.length`, `sepal length (cm)`) required careful inspection of the DataFrame on first load and a multi-branch mapping strategy.

Resolution: The `_normalise_columns()` function checks for three known schemas in order and falls back to positional assignment for unrecognised formats.

15.2 NumPy Version Conflicts

The HuggingFace `datasets` library and SHAP have conflicting NumPy version requirements. Installing them in arbitrary order caused `ValueError` and `ImportError` at runtime. The issue manifested as silent failures in SHAP’s KernelExplainer when NumPy was downgraded post-install.

Resolution: NumPy was pinned last in the installation sequence, and `requirements.txt` specifies `numpy>=1.26.0,<2.0.0` to constrain the acceptable range.

15.3 SVM Probability Calibration Cost

Setting `probability=True` on SVC triggers Platt scaling via internal cross-validation during `.fit()`, substantially increasing training time. For the Iris dataset this is negligible, but it introduced

unexpected delays during development when testing with larger synthetic datasets.

Resolution: Documented as a known limitation. For production with large datasets, `CalibratedClassifierCV` with `cv='prefit'` is the appropriate alternative.

15.4 SHAP KernelExplainer Memory and Speed

KernelExplainer is computationally expensive for SVM because it is model-agnostic and must evaluate the model exponentially many times per prediction. Using the full training set as background caused notebook timeouts.

Resolution: Background reduced to 20 randomly sampled training points (`nsamples=50` perturbations per explanation). This is an approximation, documented in the model card.

15.5 ipywidgets Rendering in Exported Notebooks

`ipywidgets` renders interactively inside a running Jupyter session but does not display in static HTML or PDF exports of the notebook. Reviewers viewing the notebook on GitHub see the cell output as empty.

Resolution: A static demo cell with 5 hardcoded sample predictions was added before the widget cell, ensuring static reviewers see representative output even without running the notebook.

Chapter 16

Limitations

Known Limitations

The following limitations apply to this project. They are documented here for intellectual honesty and to frame the scope of valid conclusions.

16.1 Dataset Scale

The Iris dataset contains 150 samples — a quantity so small that a single misclassification changes accuracy by 0.67 percentage points. Performance differences between models are not statistically robust at this scale. The near-identical performance of SVM, KNN, LR, and Ensemble (all 96.67%) should be interpreted as “all are effective” rather than evidence of superiority for any particular algorithm.

16.2 KNN Inference Complexity

KNN does not learn a parametric model. At inference time, it computes the distance between the query point and all N training points — $O(N \cdot d)$ complexity. This is acceptable for 120 training samples but becomes infeasible at industrial scale without approximate nearest-neighbour indices (FAISS, Annoy, ScaNN).

16.3 Benchmark vs Real-World Gap

The Iris dataset is artificially clean: perfectly balanced classes, no missing values, no outliers of consequence, and at least one class (Setosa) that is trivially linearly separable. Real classification tasks exhibit noise, imbalance, and overlapping distributions far more extreme than those observed here. Performance metrics achieved on Iris do not generalise as predictions of performance on real datasets.

16.4 GridSearchCV Coverage

The hyperparameter search uses a fixed, manually defined grid. This approach misses potentially better configurations outside the grid and is computationally wasteful for small gains. Bayesian optimisation (Optuna, Ray Tune) would provide more efficient coverage of a larger hyperparameter space.

16.5 SHAP Approximation

The KernelExplainer SHAP values are computed with a background of 20 samples and `nsamples=50` perturbations per data point. This is a computational approximation of the exact Shapley values. The approximation error is small for well-separated classes but may be non-trivial near decision boundaries.

Chapter 17

Conclusion

This project has successfully implemented a complete, production-grade supervised learning pipeline for multi-class classification as part of the DecodeLabs AI Industrial Training Program. All four formal specification requirements have been satisfied, and the implementation substantially exceeds the minimum specification in depth, engineering rigour, and portfolio quality.

17.1 Key Findings

1. **SVM with RBF kernel achieves the highest performance:** Accuracy 96.67%, F1 0.9666, ROC-AUC 0.9967 on the test set, and the highest cross-validation mean F1 with the lowest standard deviation — indicating both superior performance and lower generalisation variance.
2. **Petal dimensions are the dominant features:** Both the Decision Tree structure and SHAP analysis confirm that petal length and petal width are the primary discriminating features. Sepal width contributes negligibly to classification accuracy.
3. **Linear separability is class-dependent:** PCA and t-SNE projections confirm that Setosa is perfectly linearly separable from the other two species. The Versicolor–Virginica boundary is non-linear, which explains why the RBF kernel outperforms linear models when evaluated on cross-validation variance.
4. **Accuracy alone is an insufficient metric:** The class imbalance demonstration showed that a model predicting only the majority class can achieve high accuracy while F1 collapses — validating the use of macro F1 as the primary evaluation metric throughout this project.

5. **The sklearn Pipeline is the correct production unit:**

Encapsulating preprocessing and inference in a single serialisable object eliminates the most common source of production ML bugs: inconsistent feature scaling between training and inference.

17.2 Lessons Learned

The most valuable engineering lesson from this project is that *correct process matters more than model choice on clean data*. On the Iris benchmark, the choice of algorithm is almost irrelevant — what distinguishes a professional implementation is the absence of data leakage, the use of stratified splits, the proper estimation of generalisation error through cross-validation, and the discipline to persist models in a reproducible format with complete experiment logging.

17.3 Bridge to Project 3

Project 2 has established the supervised learning pipeline that Project 3 will extend into deep learning and computer vision. The conceptual bridge is direct: Project 1 used exact dictionary lookup (discrete matching); Project 2 uses learned statistical decision boundaries on tabular features; Project 3 will use convolutional neural networks to learn hierarchical feature representations directly from pixel data — extending the same IPO framework to a fundamentally higher-dimensional input space.

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